-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

Location shifted

Variation 1

-1 -1 1 -1 -1
-1 1 -1 1 -1
-1 1 1 -1 -1
-1 -1 1 -1 -1
-1 -1 1 -1 -1
-1 -1 1 -1 -1
-1 -1 1 -1 -1

→

-1 -1 1 -1 -1
-1 1 -1 1 -1
-1 1 1 -1 -1
-1 -1 1 -1 -1
-1 -1 1 -1 -1
-1 -1 1 -1 -1
-1 -1 1 -1 -1

✖

-1 1 1 1 -1
-1 1 -1 1 -1
-1 1 1 1 -1
-1 -1 -1 1 -1
-1 -1 -1 1 -1
-1 -1 1 -1 -1
-1 1 -1 -1 -1

To handle variety in digits we can use simple artificial neural network (ANN)

Image size = 1920 x 1080 X 3

Image size = 1920 x 1080 X 3

First layer neurons = 1920 x 1080 X 3 ~ 6 million

Hidden layer neurons = Let's say you keep it ~ 4 million

Weights between input and hidden layer = 6 mil * 4 mil
= 24 million

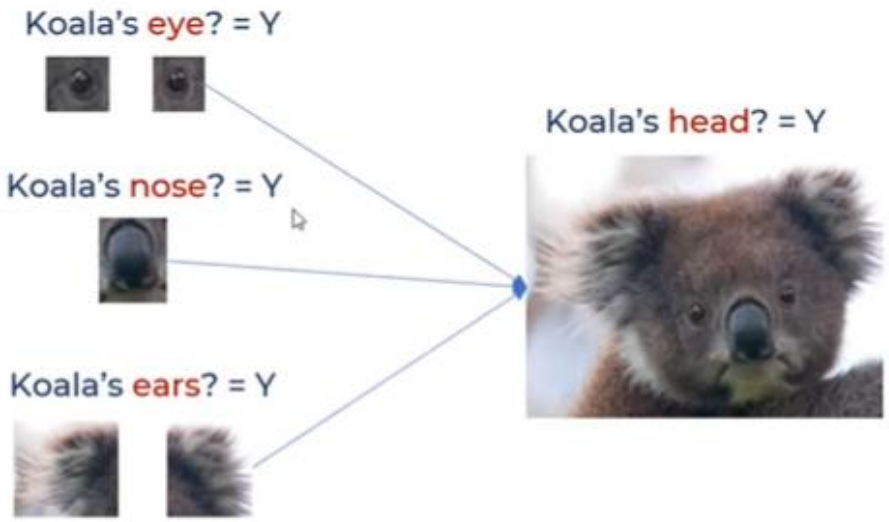
Disadvantages of using ANN for image classification

1. Too much computation
2. Treats local pixels same as pixels far apart
3. Sensitive to location of an object in an image

Koala's eye? = Y

Koala's nose? = Y

Koala's ears? = Y


Koala's eye? = Y
Koala's nose? = Y
Koala's ears? = Y
Koala's head? = Y

Koala's eye? = Y

Koala's nose? = Y

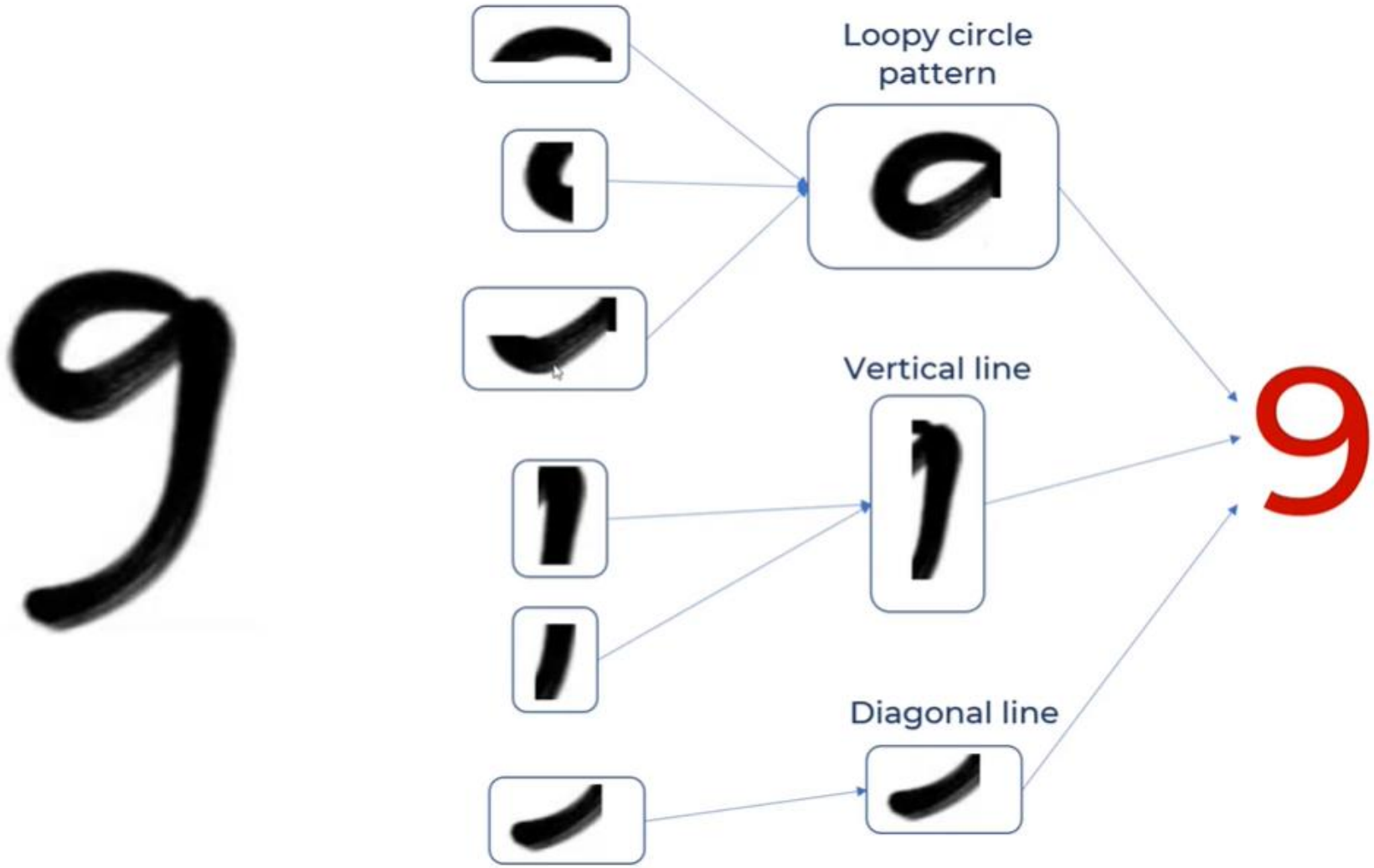
Koala's ears? = Y

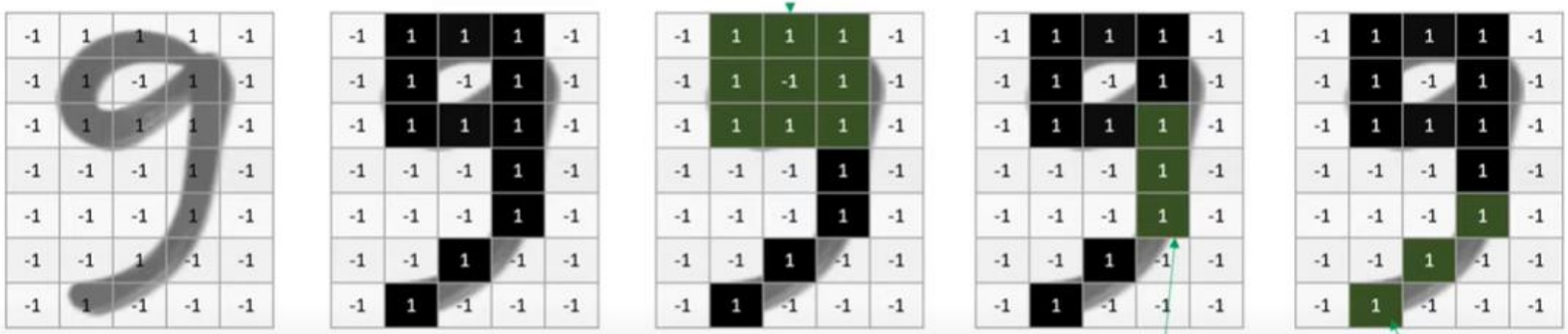
Koala's head? = Y

Koala's hands? = Y

Koala's legs? = Y

Koala's body? = Y


Loopy circle pattern
Vertical line
Diagonal line
9


Loopy pattern filter
Vertical line filter
Diagonal line filter

-1	**1**	**1**	**1**	-1
-1	**1**	-1	**1**	-1
-1	**1**	**1**	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	**1**	-1	-1
-1	**1**	-1	-1	-1

1	1	1
1	-1	1
1	1	1

$$-1+1+1-1-1-1-1+1+1 = -1 \rightarrow -1/9 = -0.11$$

-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

*

1	1	1
1	-1	1
1	1	1

-0.11		

-1	**1**	**1**	**1**	-1
-1	**1**	-1	**1**	-1
-1	**1**	**1**	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	**1**	-1	-1
-1	**1**	-1	-1	-1

*

1	1	1
1	-1	1
1	1	1

-0.11	**1**	

-1	**1**	**1**	**1**	-1
-1	**1**	-1	**1**	-1
-1	**1**	**1**	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	**1**	-1	-1
-1	**1**	-1	-1	-1

*

1	1	1
1	-1	1
1	1	1

-0.11	1	-0.11

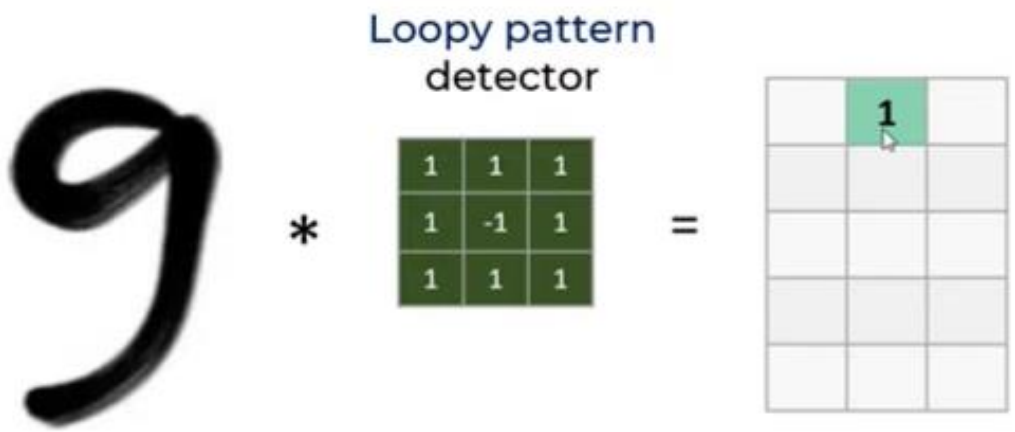
-1	**1**	**1**	**1**	-1
-1	**1**	-1	**1**	-1
-1	**1**	**1**	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	**1**	-1	-1
-1	**1**	-1	-1	-1

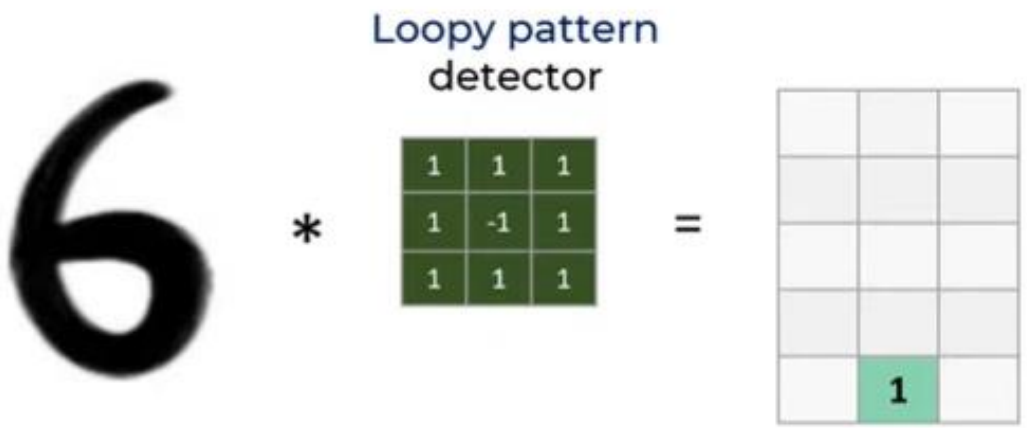
*

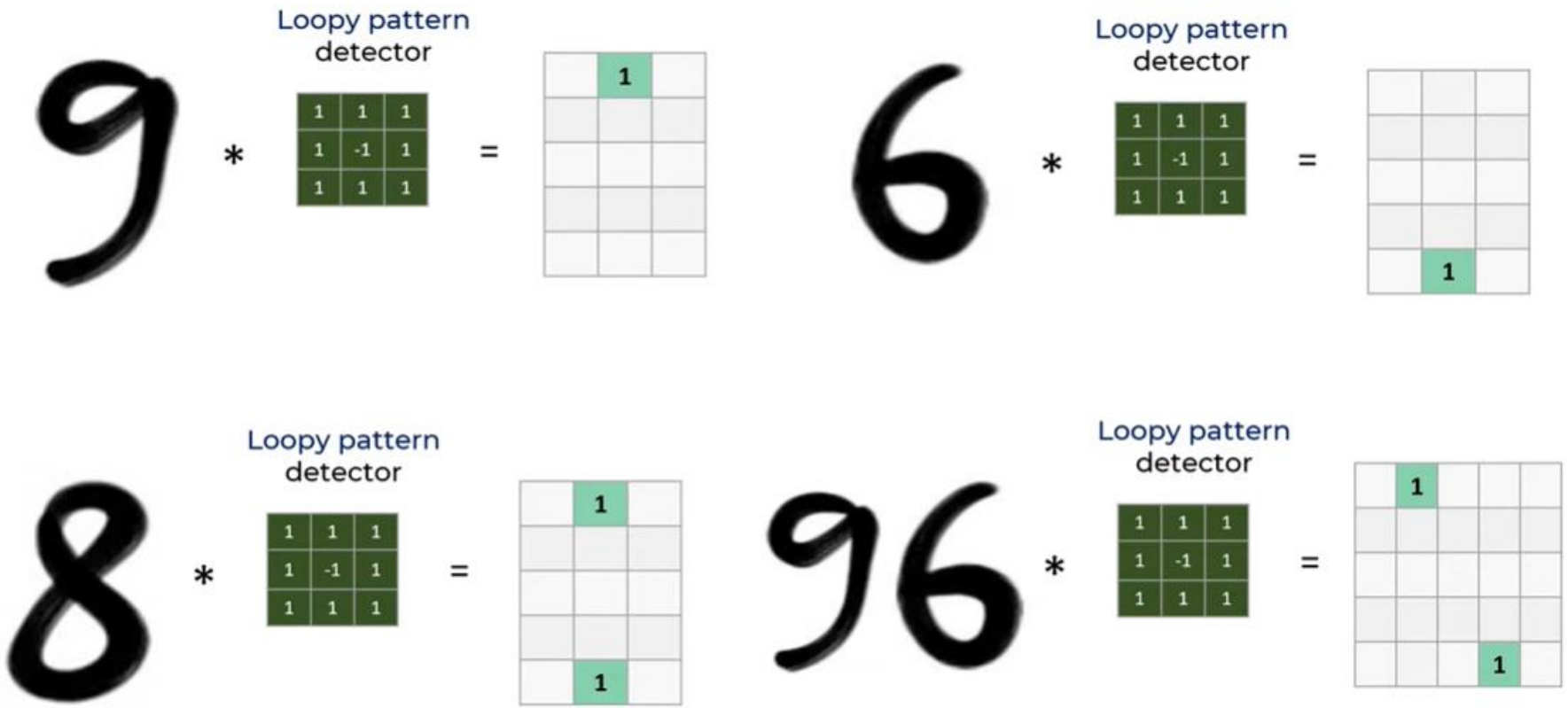
1	1	1
1	-1	1
1	1	1

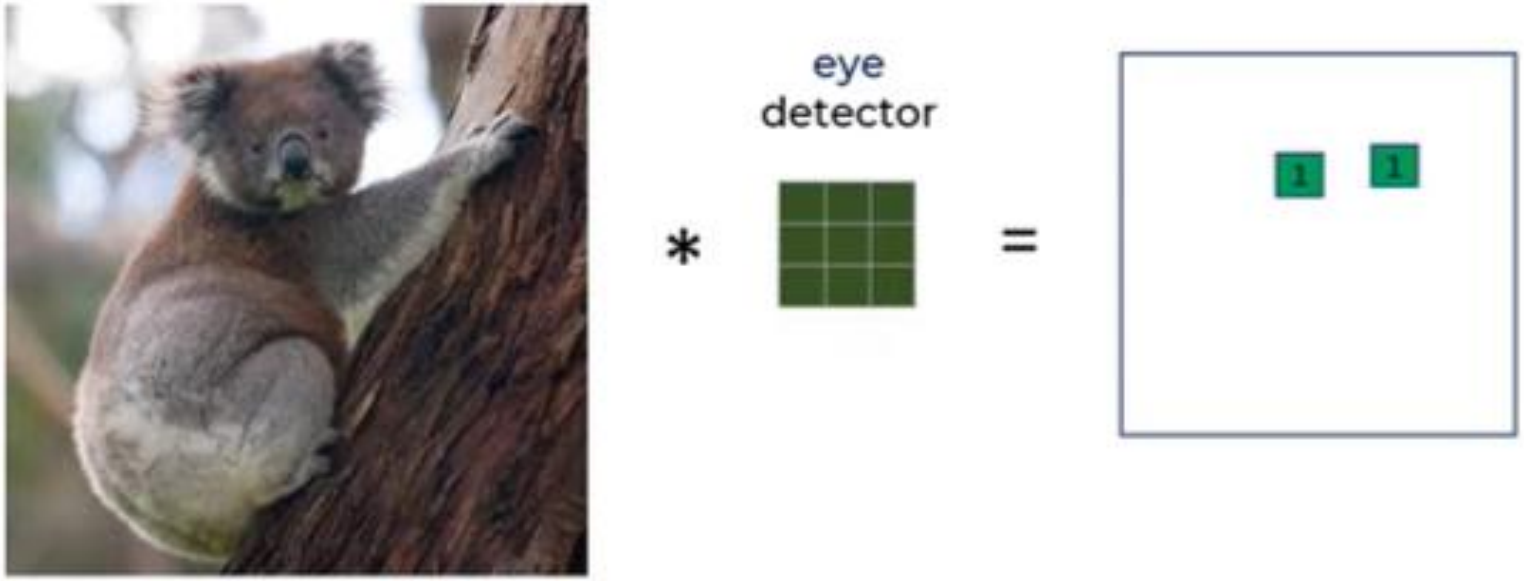
-0.11	**1**	-0.11
-0.55	0.11	-0.33
-0.33	0.33	-0.33
-0.22	-0.11	-0.22
-0.33	-0.33	-0.33

Feature Map


Loopy pattern detector
1 1 1
1 -1 1
1 1 1
1


Loopy pattern detector
1 1 1
1 -1 1
1 1 1
1

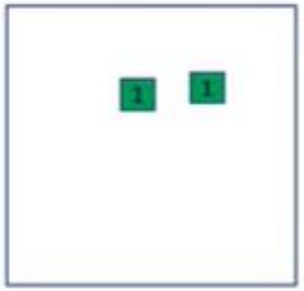

Loopy pattern detector
1 1 1
1 -1 1
1 1 1
1
Loopy pattern detector
1 1 1
1 -1 1
1 1 1
1
Loopy pattern detector
1 1 1
1 -1 1
1 1 1
1
1
Loopy pattern detector
1 1 1
1 -1 1
1 1 1
1
1


eye
detector
*
=
1
1

*

eye
detector

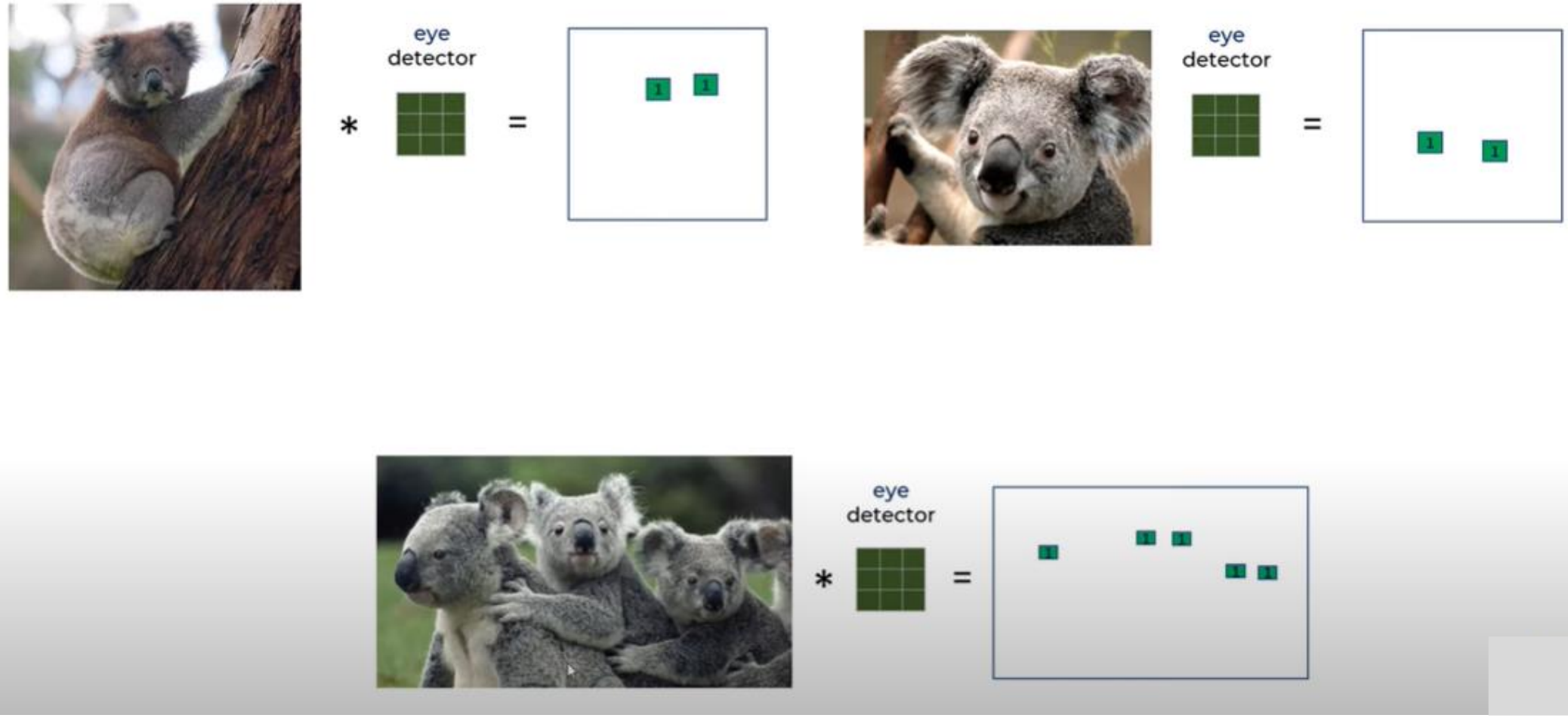
=

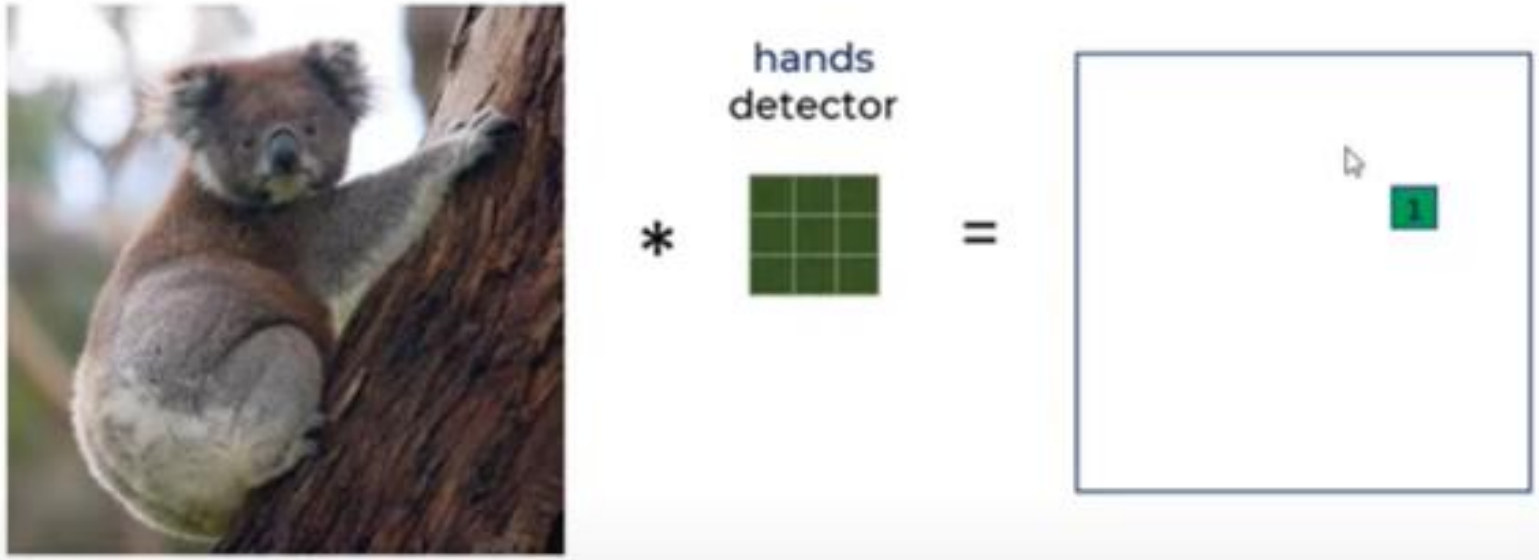


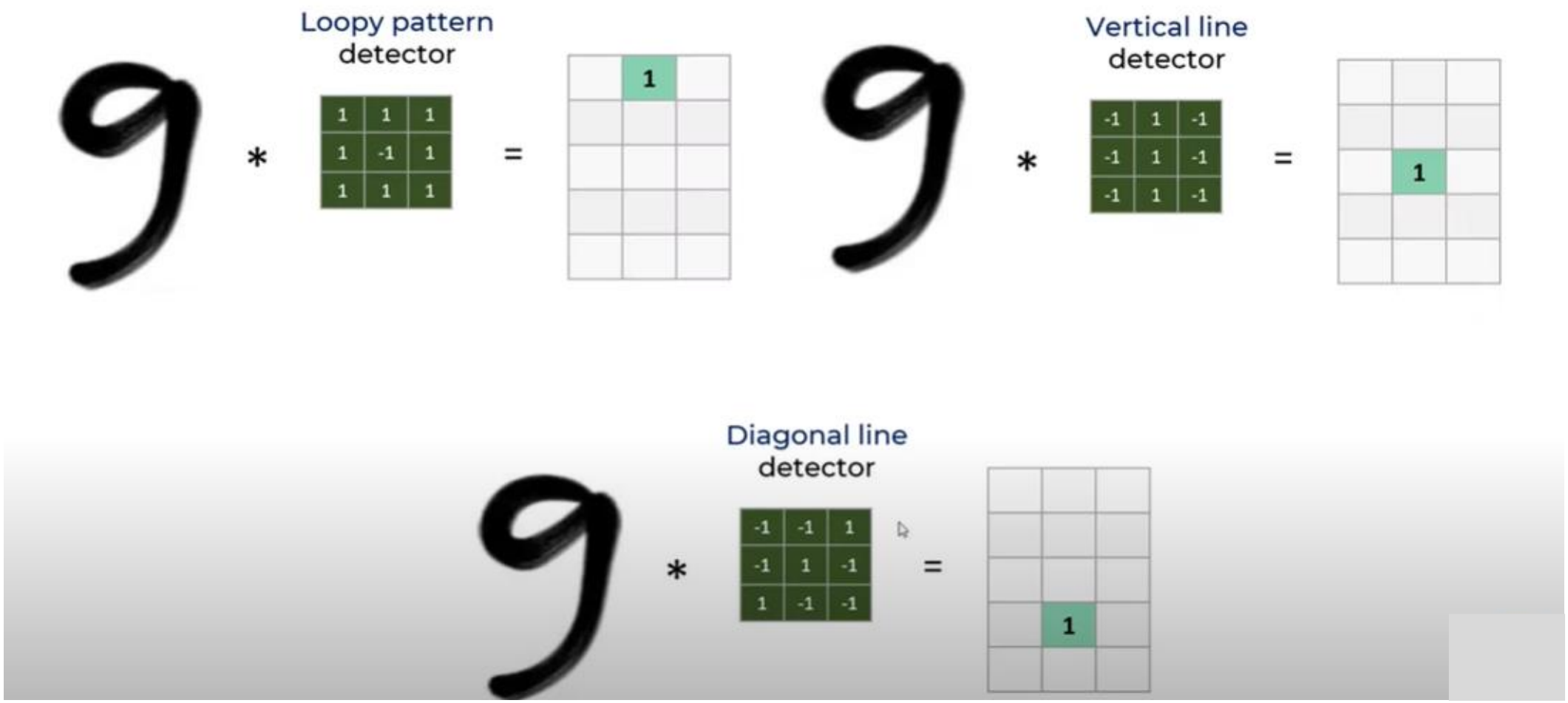
eye
detector

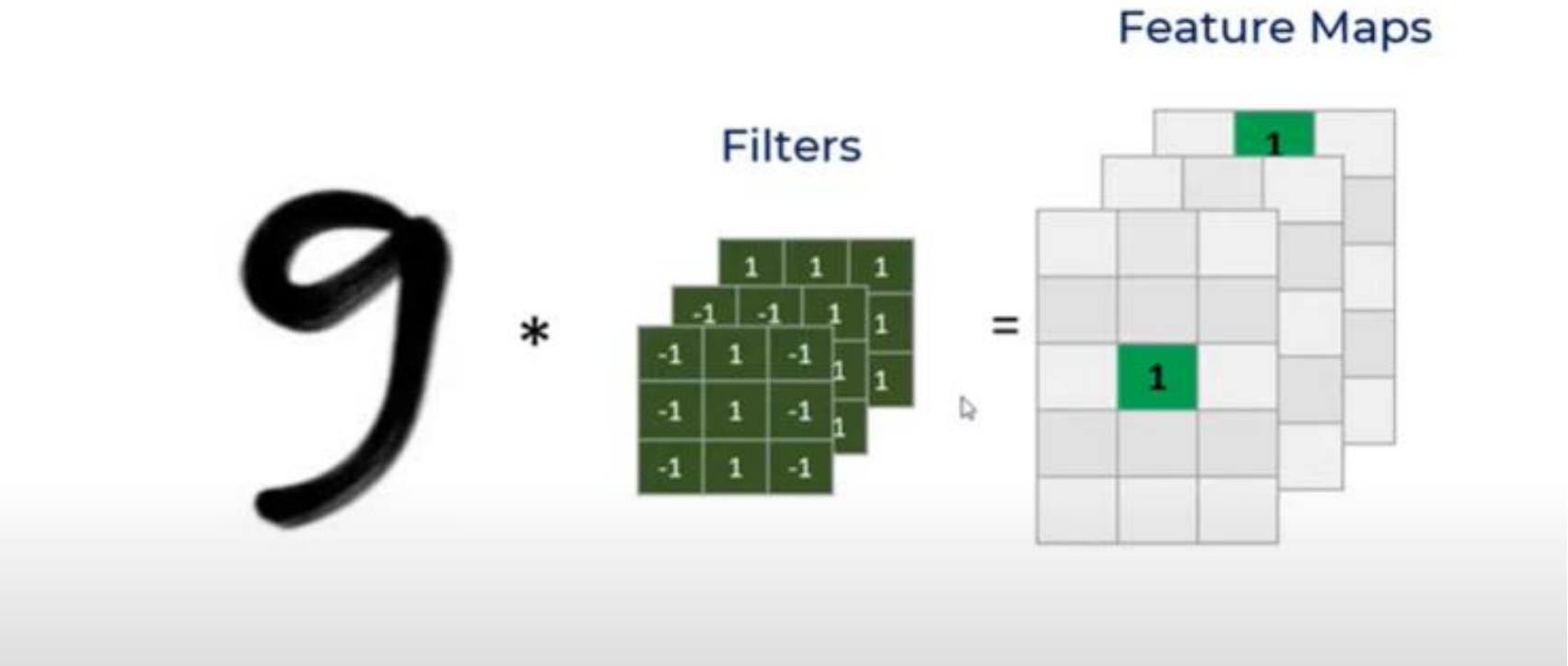
=

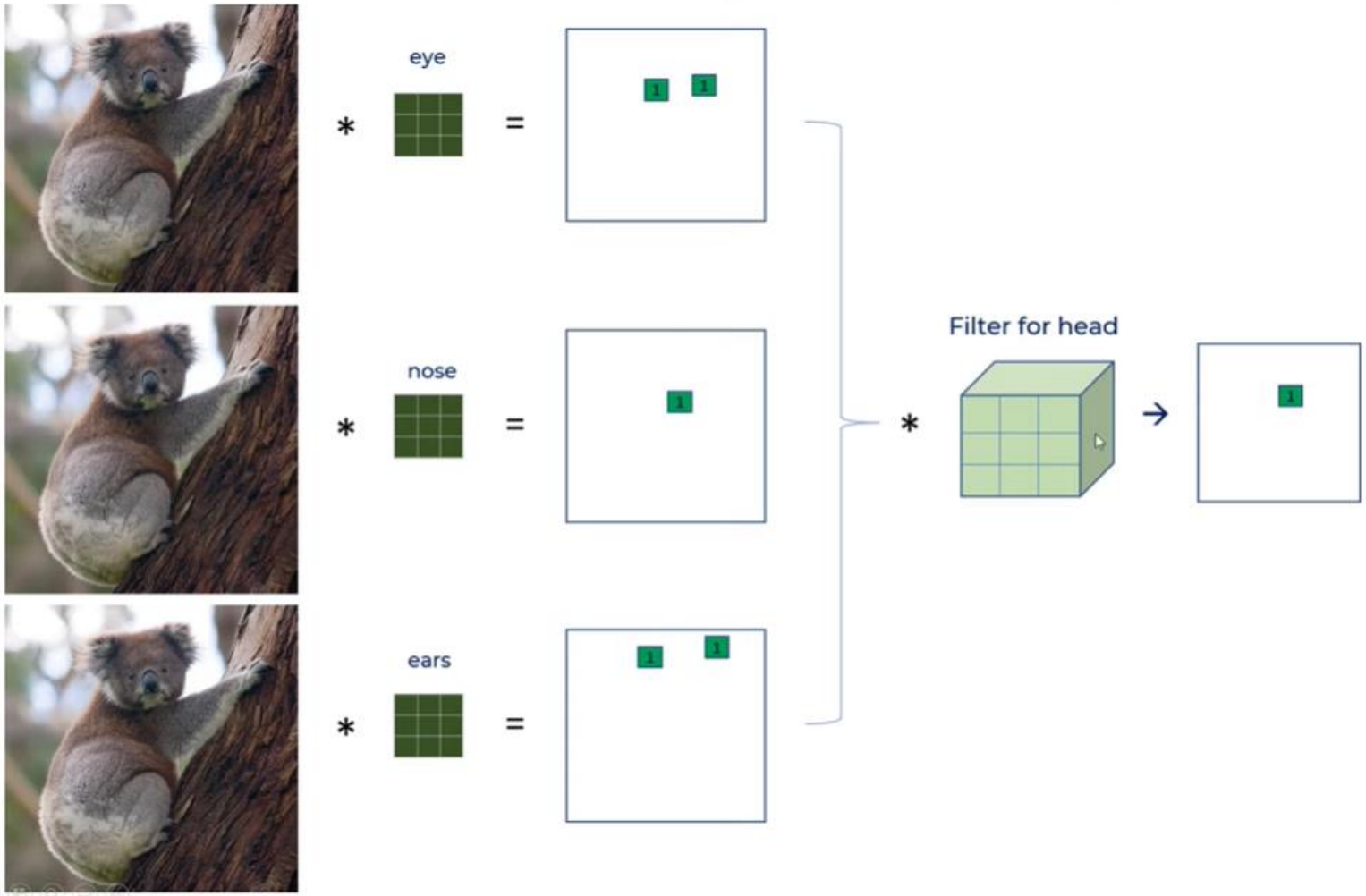
Location invariant: It can detect eyes in any location of the image

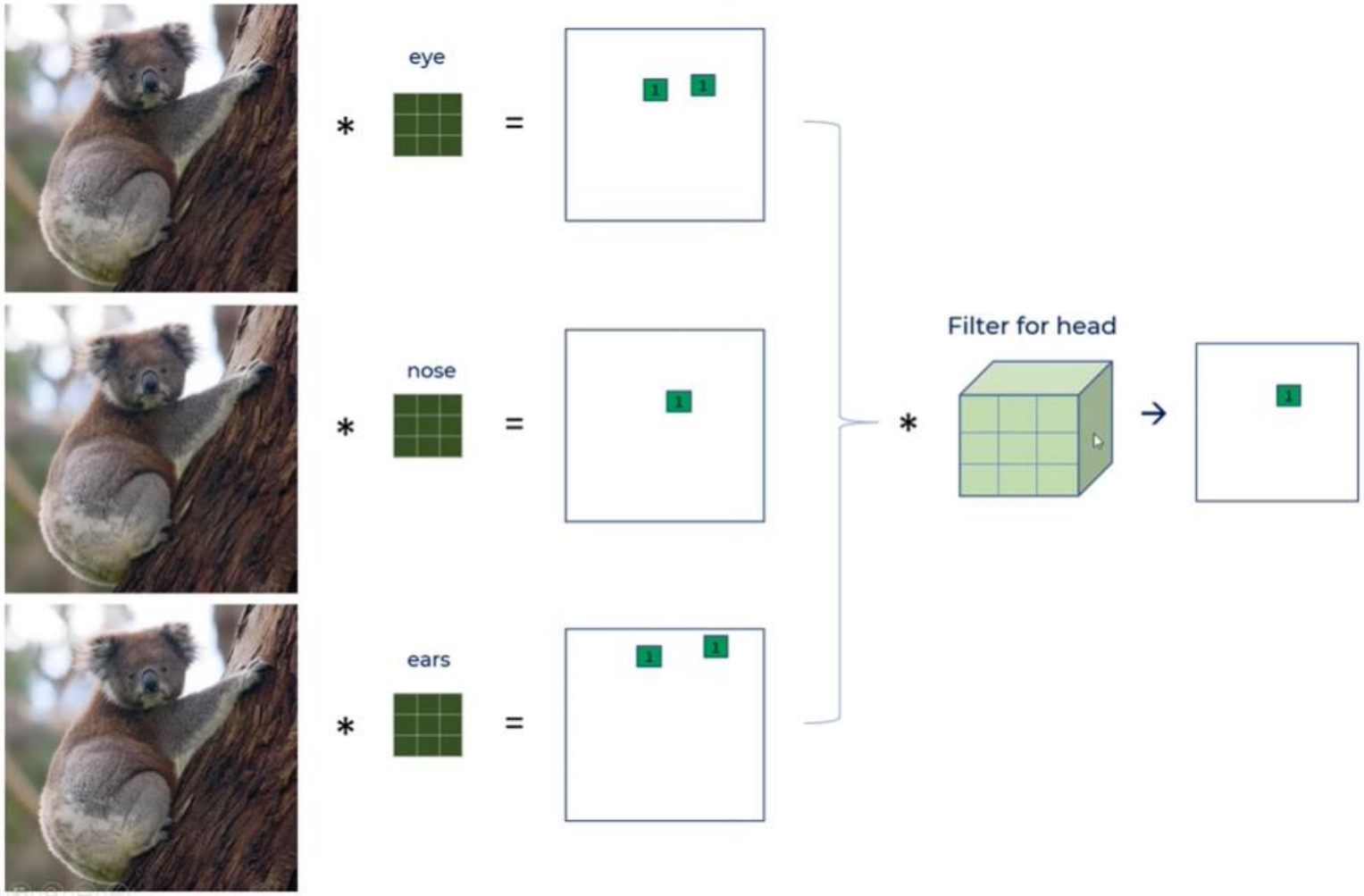


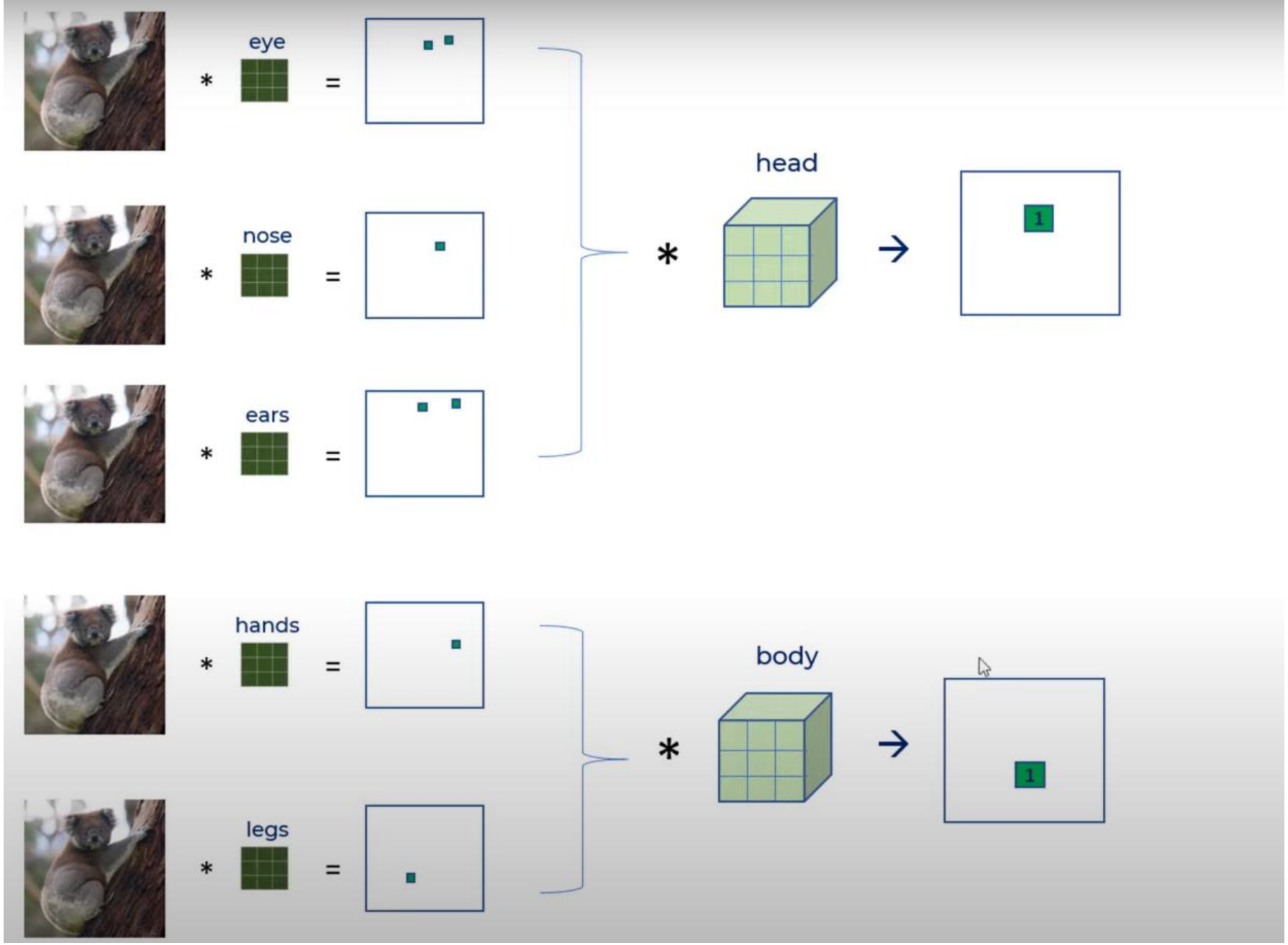

hands
detector
*
=
1

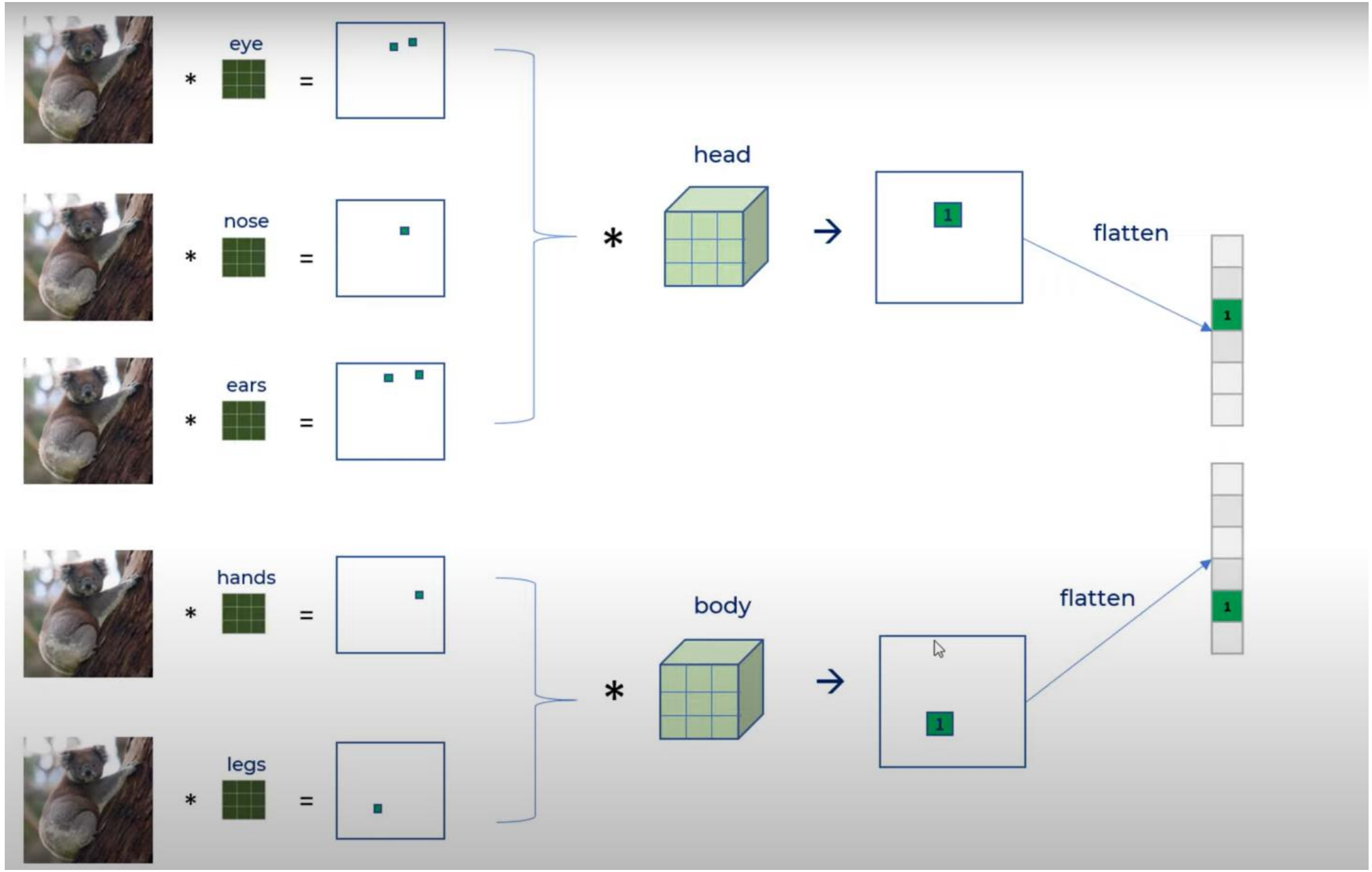

Loopy pattern detector
1 1 1
1 -1 1
1 1 1
1
Vertical line detector
-1 1 -1
-1 1 -1
-1 1 -1
1
Diagonal line detector
-1 -1 1
-1 1 -1
1 -1 -1
1

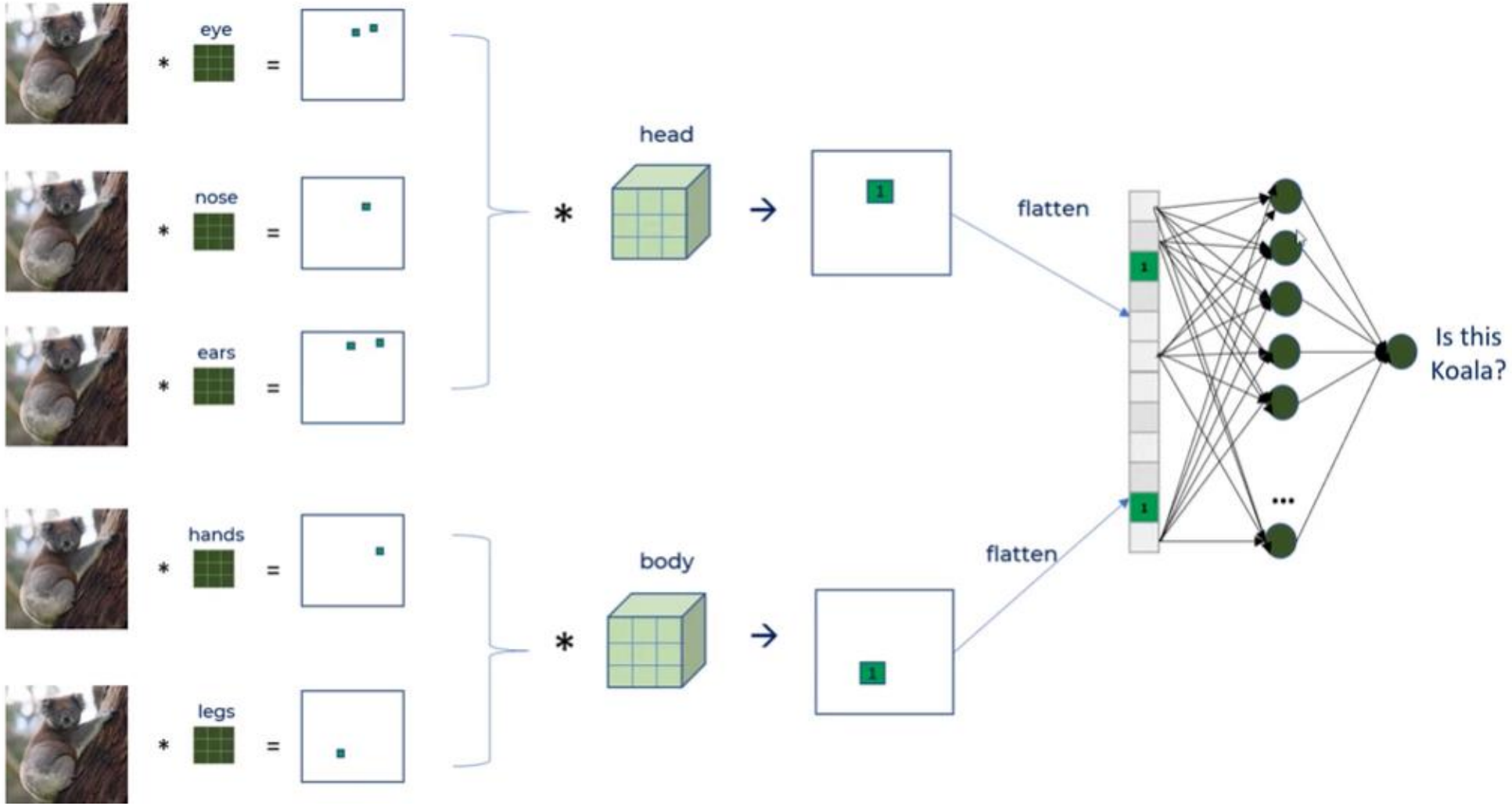

Feature Maps
Filters
*
=
1
1 1 1
-1 -1 1
-1 1 -1
-1 1 -1
-1 1 -1
1

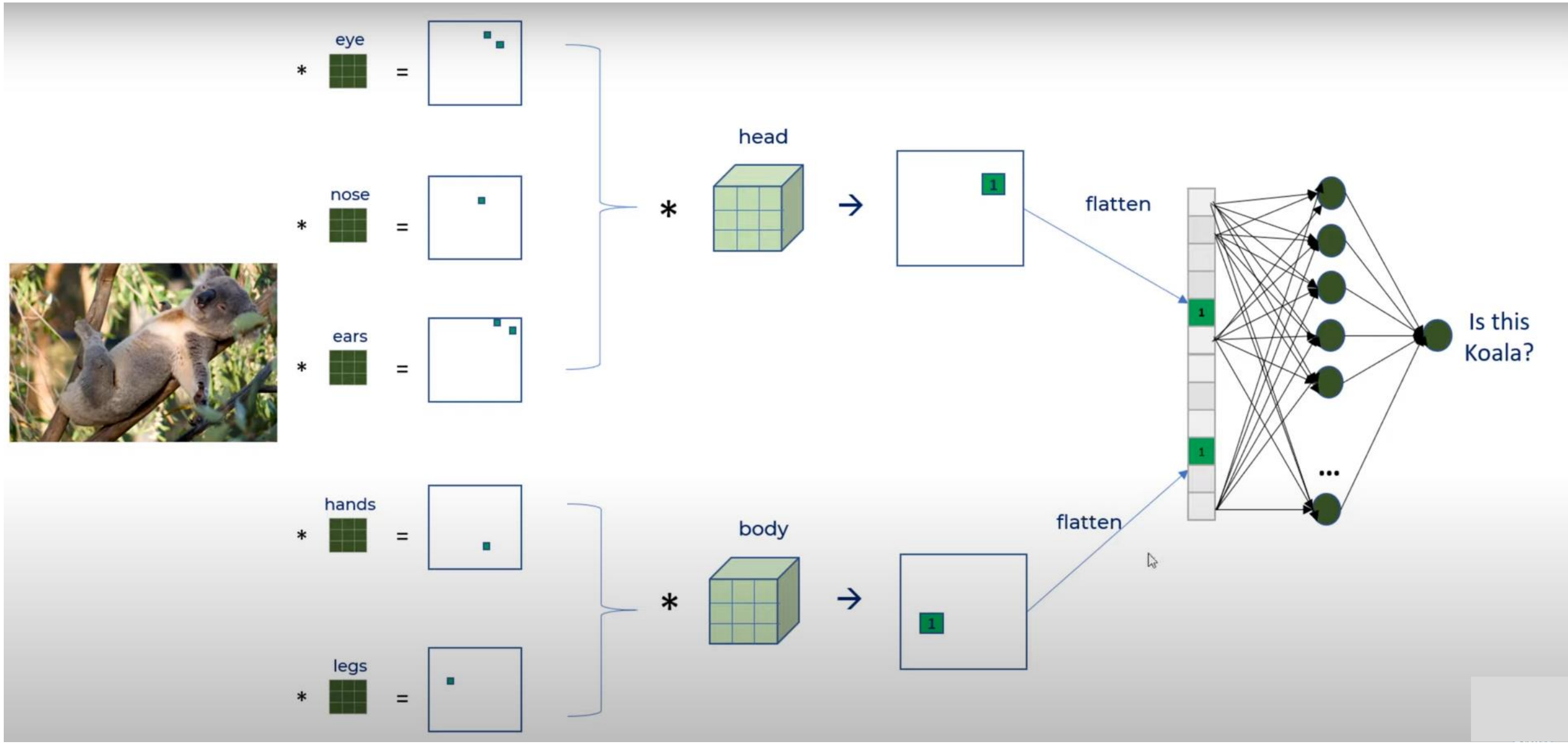

eye
nose
ears
Filter for head

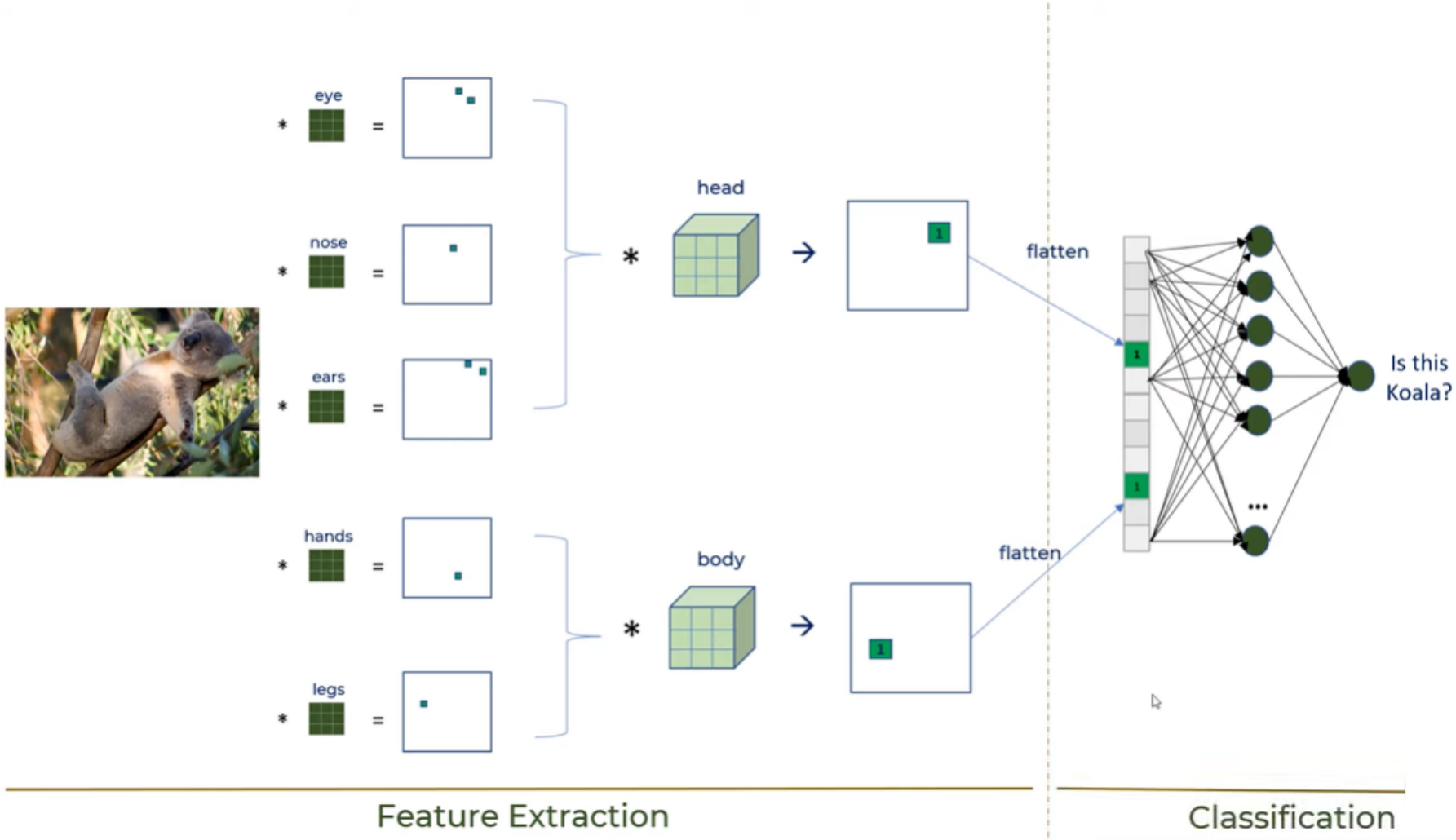

eye
nose
ears
Filter for head


eye
nose
ears
head
1
hands
legs
body
1

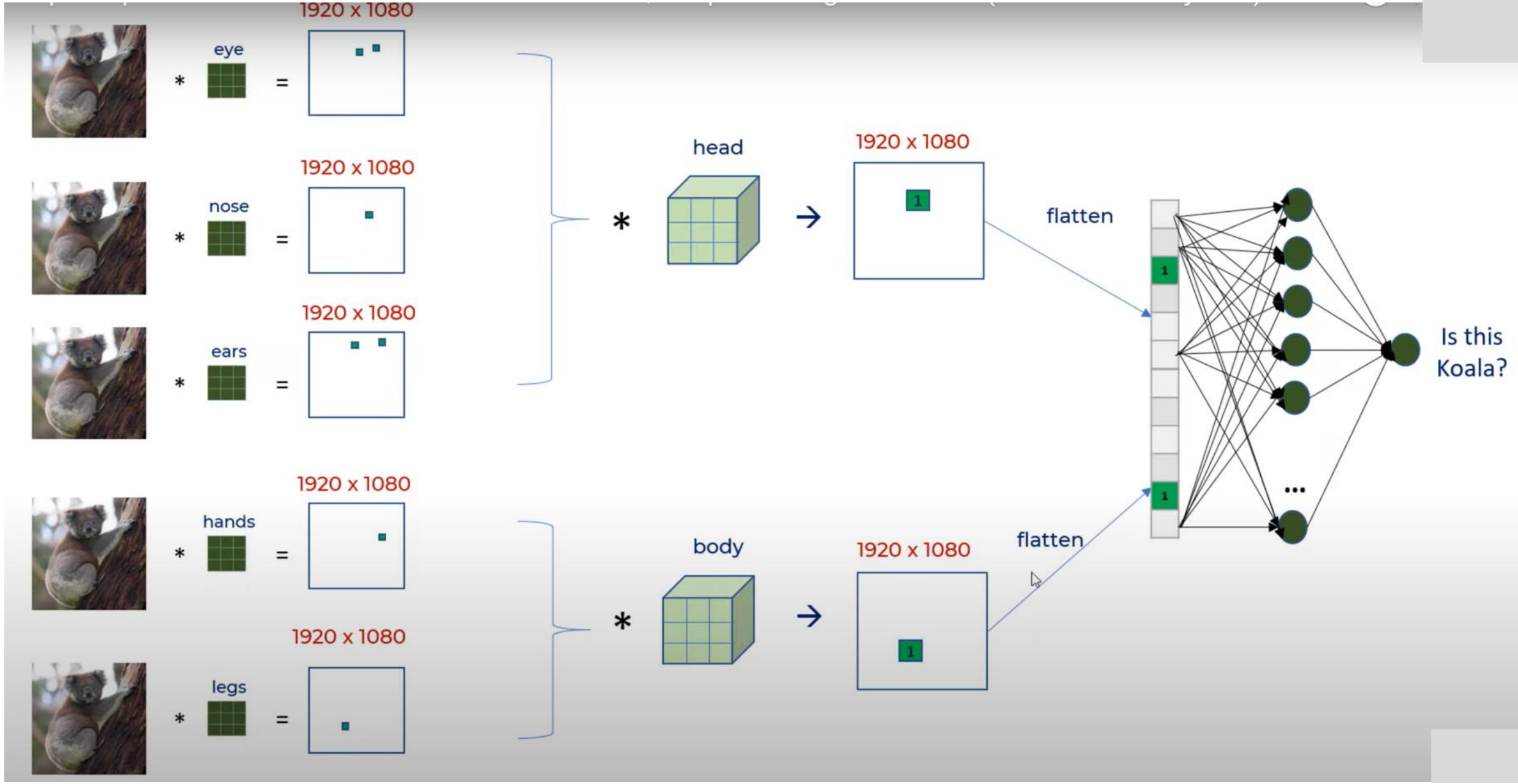

eye
nose
ears
hands
legs
head
body
flatten
flatten
1
1
1


eye
nose
ears
hands
legs
head
body
flatten
flatten
Is this Koala?


eye
nose
ears
hands
legs
head
body
flatten
flatten
Is this Koala?


eye
nose
ears
hands
legs
head
body
flatten
flatten
Is this Koala?
Feature Extraction
Classification

ReLU helps with making the model nonlinear


1920 x 1080
eye
*
=
1920 x 1080
nose
*
=
1920 x 1080
ears
*
=
1920 x 1080
hands
*
=
1920 x 1080
legs
*
=
head
*
1920 x 1080
1
flatten
body
*
1920 x 1080
1
flatten
1
1
...
Is this Koala?

-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

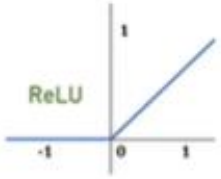
*

Loopy pattern filter

1	1	1
1	-1	1
1	1	1

→

-0.11	1	-0.11
-0.55	0.11	-0.33
-0.33	0.33	-0.33
-0.22	-0.11	-0.22
-0.33	-0.33	-0.33



→

0	1	0
0	0.11	0
0	0.33	0
0	0	0
0	0	0

Pooling layer is used to reduce the size

5	1	3	4
8	2	9	2
1	3	0	1
2	2	2	0

8	**9**
3	2

2 by 2 filter with stride = 2

Loopy pattern filter
ReLU
Max pooling

0	**1**	0
0	0.11	0
0	0.33	0
0	0	0
0	0	0

1	**1**
0.33	0.33
0.33	0.33
0	0

2 by 2 filter with stride = 1

Shifted 9 at different position

1	1	1	-1	-1
1	-1	1	-1	-1
1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1
1	-1	-1	-1	-1

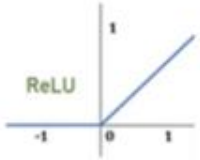
*

Loopy pattern filter

1	1	1
1	-1	1
1	1	1

→

1	-0.11	-0.11
0.11	-0.33	0.33
0.33	-0.33	-0.33
-0.11	-0.55	-0.33
-0.55	-0.33	-0.55



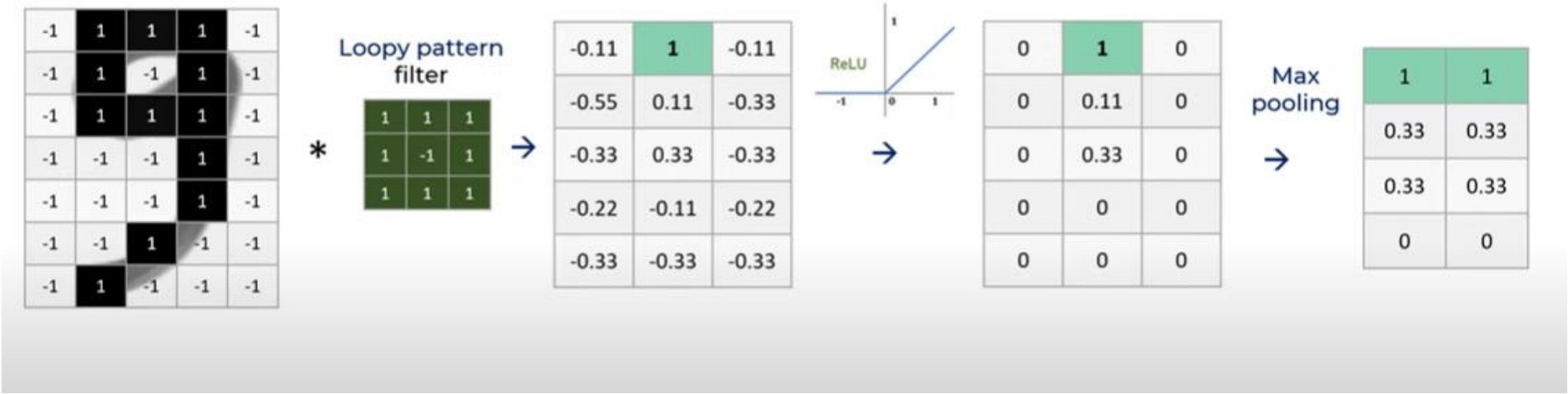
→

1	0	0
0.11	0	0.33
0.33	0	0
0	0	0
0	0	0

Max pooling

→

1	0.33
0.33	0.33
0.33	0
0	0


Loopy pattern filter
ReLU
Max pooling

Shifted 9 at different position

1	1	1	-1	-1
1	-1	1	-1	-1
1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1
1	-1	-1	-1	-1

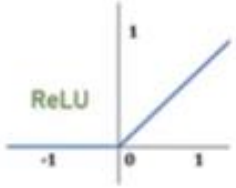
*

Loopy pattern filter

1	1	1
1	-1	1
1	1	1

→

1	-0.11	-0.11
0.11	-0.33	0.33
0.33	-0.33	-0.33
-0.11	-0.55	-0.33
-0.55	-0.33	-0.55



→

1	0	0
0.11	0	0.33
0.33	0	0
0	0	0
0	0	0

Max pooling

→

1	0.33
0.33	0.33
0.33	0
0	0

There is average pooling also...

5	1	3	4
8	2	9	2
1	3	0	1
2	2	2	0

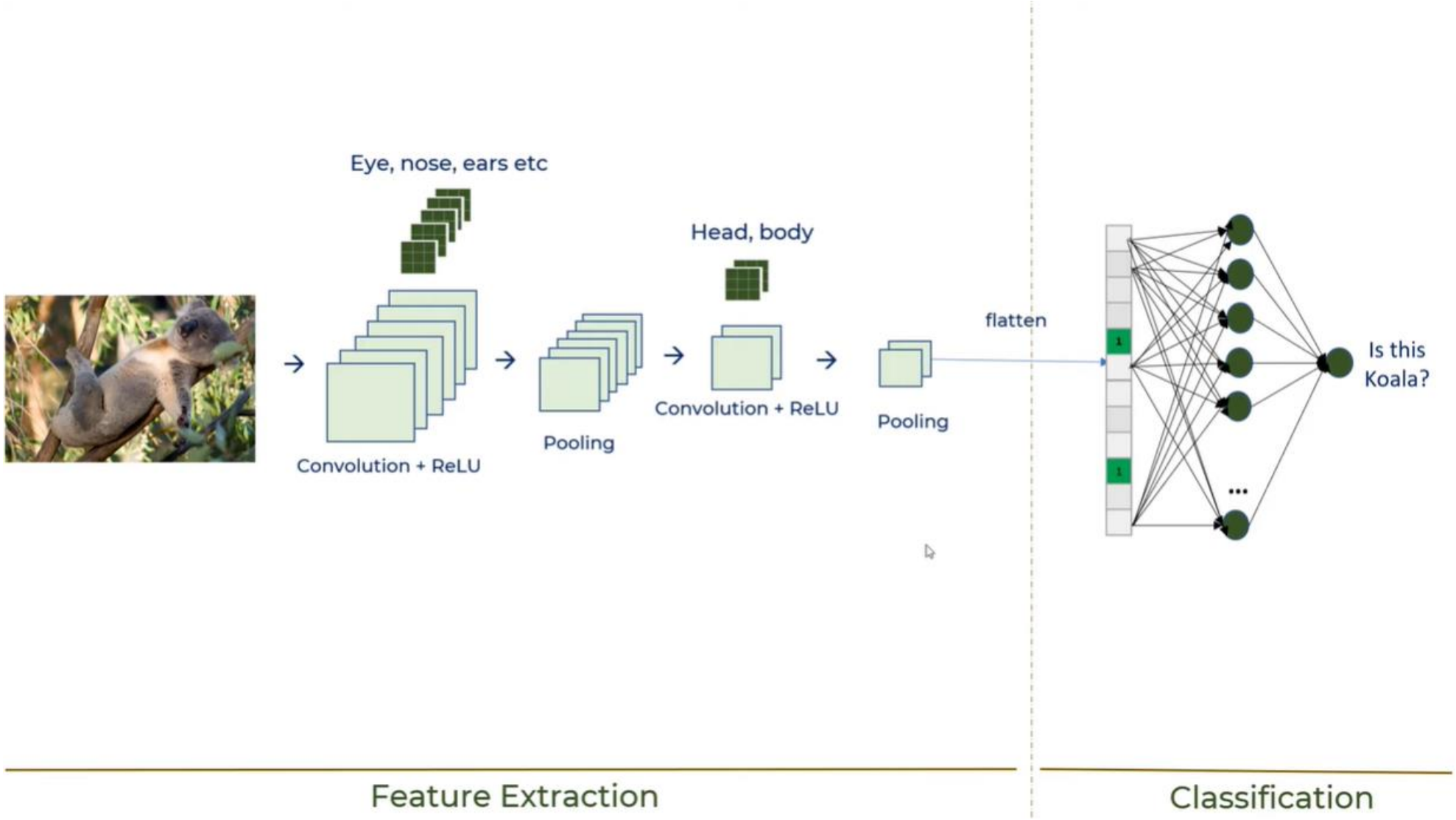
4	4.5
2	0.75

Benefits of pooling

Reduces dimensions & computation

Reduce overfitting as there are less parameters

Model is tolerant towards variations, distortions


Eye, nose, ears etc
Head, body
Convolution + ReLU
Pooling
Convolution + ReLU
Pooling
flatten
Is this Koala?
Feature Extraction
Classification

Convolution

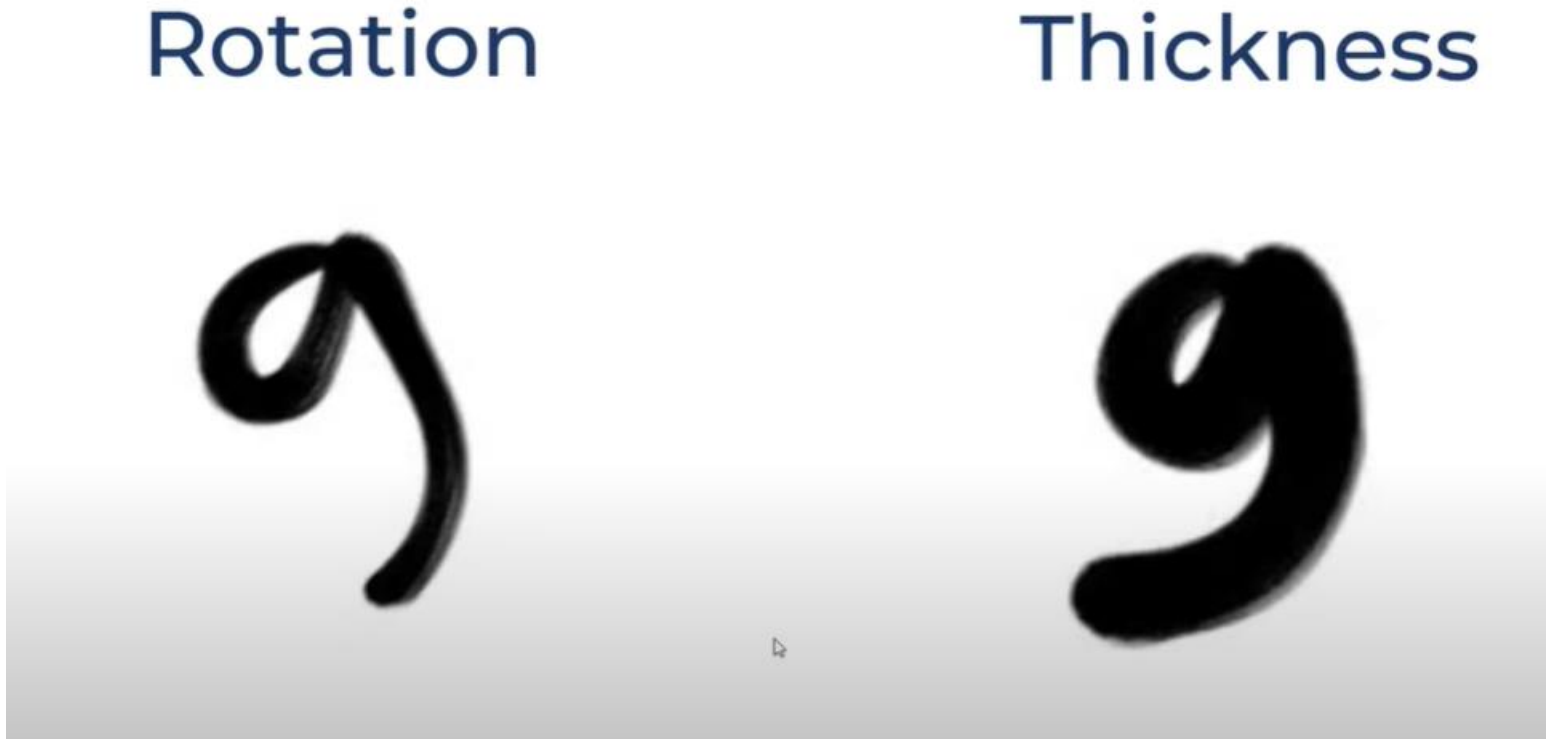
- Connections sparsity reduces overfitting
- Conv + Pooling gives location invariant feature detection
- Parameter sharing

ReLU

- Introduces nonlinearity
- Speeds up training, faster to compute

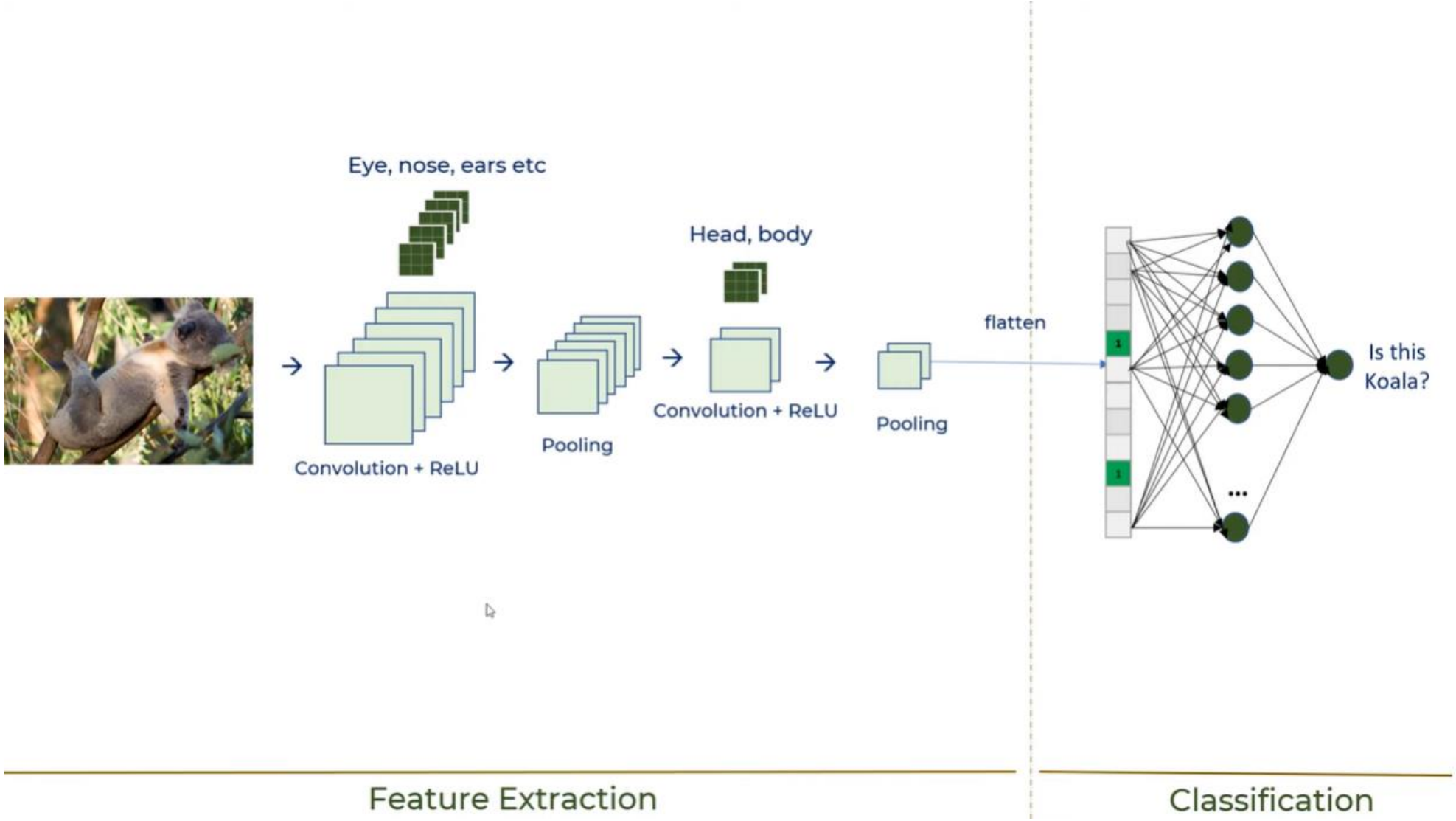
Pooling

- Reduces dimensions and computation
- Reduces overfitting
- Makes the model tolerant towards small distortion and variations


Rotation
Thickness

CNN by itself doesn't take care of rotation and scale

- You need to have rotated, scaled samples in training dataset
- If you don't have such samples than use data augmentation methods to generate new rotated/scaled samples from existing training samples


Eye, nose, ears etc
Head, body
Convolution + ReLU
Pooling
Convolution + ReLU
Pooling
flatten
Is this Koala?
Feature Extraction
Classification